

Code No. M24PC407AI

PART-B

Answer Any Five questions. (5X8M=40M)

Q.No. Questions Marks CO BTL

2	a	What is ranking of data? Explain different ranking methods used in data analysis.	4	1	1
	b	Explain the process of reading and selecting data in Python using suitable examples.	4	1	3
3.	a	Explain data summarization techniques used in descriptive statistics.	4	2	3
	b	Explain variance and standard deviation. Why are they important in data analysis?	4	2	3
		Demonstrate the working principle of Support Vector Machines (SVM) with an example.	8	3	2
	a	Explain quality measures of clustering and describe a case study where clustering is applied.	4	4	5
	b	Evaluate different python libraries used in data science	4	4	3
		Explain the working principle of PageRank and mention its importance. —	8	5	4
7.	a	Illustrate the data plotting and visualization in Python. Describe the role of Matplotlib in data visualization.	4	1	4
	b	A sample of 36 students has mean 52 and p=6 (row) test at 5% level whether the population mean is 50	4	2	4
	a	Compare the roles of training, validation, and test datasets in supervised learning.	4	3	4
	b	What is polynomial regression? Explain when and why it is used.	4	4	2
9.	a	Compute degree centrality for graph A-B, A-C, B-C, C-D.	4	5	5
	b	Differentiate between linear and non linear SVM regression with an suitable data set.	4	2	4

Code MB24PC001A1

METHODIST COLLEGE OF ENGINEERING & TECHNOLOGY

(An Autonomous Institution)

B.E. (CSE(AD)) IV-Semester (Regular) (M24) Examination, JUNE/JULY-2020

Subject: DATA SCIENCE

Max. Marks: 60

Time: 3 hours

Note: Missing data, if any, maybe suitably assumed.

PART-A

Answer All the questions. (10X2M=20M)

Q.No.	Questions	Marks	CO	BTL
1. a	What is Python? Mention any two reasons why it is preferred by data scientists.	2	1	1
b	List the measures used in the descriptive statistics along with examples.	2	1	1
c	Define descriptive statistics. Mention any two measures used in descriptive statistics.	2	2	2
d	Write the formula for Z-score	2	2	1
e	What is supervised learning?	2	3	1
f	Define under-fitting and Over-fitting of a model.	2	3	2
g	State regression analysis?	2	4	1
h	What is meant by clustering in machine learning?	2	4	1
i	Define social network?	2	5	1
j	What is an ego-network?	2	5	1

PTO